

**Domain mapping:** 0 = Child, 1 = College, 2 = Social. Within each column, the best result is bold and the second-best result is underlined.

**Table 1.** Out-of-domain accuracy (%) for seed 0. Results use IID model selection.

| Algorithm | Child         | College       | Social        | Avg.          |
|-----------|---------------|---------------|---------------|---------------|
| ERM       | <b>58.163</b> | 50.358        | 55.000        | 54.507        |
| Mixup     | <u>56.463</u> | 47.733        | 54.545        | 52.914        |
| SagNet    | 52.381        | 46.301        | 57.727        | 52.136        |
| Fish      | 53.741        | <u>52.267</u> | <u>58.182</u> | <u>54.730</u> |
| SWAD      | <b>58.163</b> | 46.778        | <b>59.091</b> | 54.677        |
| Arith     | 54.762        | <b>53.938</b> | 56.364        | <b>55.021</b> |
| GGA       | 53.741        | 46.539        | 56.364        | 52.215        |
| ALOFT-S   | 55.102        | 41.527        | 53.182        | 49.937        |
| ALOFT-E   | 54.762        | 51.074        | <u>58.182</u> | 54.673        |

**Table 2.** Out-of-domain accuracy (%) for seed 1. Results use IID model selection.

| Algorithm | Child         | College       | Social        | Avg.          |
|-----------|---------------|---------------|---------------|---------------|
| ERM       | 50.680        | 55.609        | 54.545        | 53.611        |
| Mixup     | <u>52.381</u> | 52.267        | 55.455        | 53.368        |
| SagNet    | 51.701        | 50.358        | <b>61.818</b> | <u>54.626</u> |
| Fish      | 51.701        | 49.642        | 54.545        | 51.963        |
| SWAD      | 50.340        | <u>56.086</u> | 57.273        | 54.566        |
| Arith     | <b>55.102</b> | 48.449        | <u>57.727</u> | 53.759        |
| GGA       | 51.020        | 50.119        | 57.273        | 52.804        |
| ALOFT-S   | 46.259        | 50.119        | 57.273        | 51.217        |
| ALOFT-E   | 54.762        | <b>56.802</b> | 54.091        | <b>55.218</b> |

**Table 3.** Out-of-domain accuracy (%) for seed 2. Results use IID model selection.

| Algorithm | Child         | College       | Social        | Avg.          |
|-----------|---------------|---------------|---------------|---------------|
| ERM       | <u>54.082</u> | <b>60.143</b> | 58.636        | <u>57.620</u> |
| Mixup     | 53.401        | 53.222        | 54.545        | 53.723        |
| SagNet    | 52.721        | 51.790        | 56.364        | 53.625        |
| Fish      | 53.741        | 49.642        | 60.000        | 54.461        |
| SWAD      | 51.020        | 43.437        | 54.091        | 49.516        |
| Arith     | 52.381        | 46.778        | <b>62.727</b> | 53.962        |
| GGA       | <b>58.163</b> | <u>56.086</u> | 58.636        | <b>57.628</b> |
| ALOFT-S   | 53.401        | 51.551        | <u>62.273</u> | 55.742        |
| ALOFT-E   | 54.422        | 51.551        | 55.909        | 53.961        |

**Table 4.** Out-of-domain accuracy (%) for seed 3. Results use IID model selection.

| Algorithm | Child         | College       | Social        | Avg.          |
|-----------|---------------|---------------|---------------|---------------|
| ERM       | 54.082        | 52.029        | 54.091        | 53.400        |
| Mixup     | 51.701        | 53.699        | 50.909        | 52.103        |
| SagNet    | <b>54.762</b> | 50.119        | 44.091        | 49.657        |
| Fish      | <u>54.422</u> | <b>56.086</b> | 55.909        | <u>55.472</u> |
| SWAD      | 54.082        | 53.699        | <u>56.818</u> | 54.866        |
| Arith     | 52.041        | 46.062        | 50.909        | 49.671        |
| GGA       | <b>54.762</b> | <u>55.847</u> | 55.909        | <u>55.506</u> |
| ALOFT-S   | 50.340        | 58.950        | 51.364        | 53.551        |
| ALOFT-E   | 52.381        | 55.370        | <b>59.545</b> | <b>55.765</b> |

**Table 5.** Out-of-domain accuracy (%) for seed 4. Results use IID model selection.

| Algorithm | Child         | College       | Social        | Avg.          |
|-----------|---------------|---------------|---------------|---------------|
| ERM       | <u>58.503</u> | <u>57.757</u> | <u>60.000</u> | <b>58.753</b> |
| Mixup     | 57.143        | 52.745        | 54.545        | 54.811        |
| SagNet    | 56.463        | 48.210        | <u>60.000</u> | 54.891        |
| Fish      | 56.122        | 57.041        | 57.273        | 56.812        |
| SWAD      | 56.122        | 56.802        | 54.091        | 55.672        |
| Arith     | 54.762        | <b>59.427</b> | <b>60.909</b> | 58.366        |
| GGA       | <b>58.844</b> | 57.518        | 59.091        | <u>58.484</u> |
| ALOFT-S   | 55.442        | 54.177        | 55.000        | 54.873        |
| ALOFT-E   | 54.422        | 55.609        | 59.545        | 56.525        |

**Table 6.** Mean out-of-domain accuracy (%) over five seeds. Results use IID model selection.

| Algorithm | Child         | College       | Social        | Avg.          |
|-----------|---------------|---------------|---------------|---------------|
| ERM       | <u>55.102</u> | <b>55.179</b> | 56.454        | <b>55.578</b> |
| Mixup     | 54.218        | 51.933        | 54.000        | 53.384        |
| SagNet    | 53.606        | 49.356        | 56.000        | 52.987        |
| Fish      | 53.945        | 52.936        | 57.182        | 54.688        |
| SWAD      | 53.945        | 51.360        | 56.273        | 53.859        |
| Arith     | 53.810        | 50.931        | <b>57.727</b> | 54.156        |
| GGA       | <b>55.306</b> | 53.222        | <u>57.455</u> | <u>55.328</u> |
| ALOFT-S   | 52.109        | 51.265        | 55.818        | 53.064        |
| ALOFT-E   | 54.150        | <u>54.081</u> | 57.454        | 55.228        |